

Response to the Editor

Scientific Reports

Submission ID: 859df4a2-2669-4740-a5ac-4bede5db505f

Manuscript Title: *Explainable Machine Learning Analysis of Factors Associated with Skilled Birth Attendance in Burkina Faso*

Dear Dr Singh,

Thank you very much for your positive assessment of our manuscript and for the opportunity to make this minor revision. We appreciate the Editor's comments and have carefully addressed all requested points.

In the revised manuscript, the requested changes are highlighted in **green** for ease of identification. The page and line numbers provided below refer to the revised manuscript.

Editor Comment 1: Code deposition and availability

Comment:

If the manuscript uses any custom or bespoke computational tool or code, or reports a new algorithm, tool, software, or pipeline, the underlying code must be deposited in a recognised DOI-assigning repository (e.g. Zenodo) and linked either from Methods or a dedicated Code Availability section.

Response:

We thank the Editor for this important recommendation. In accordance with the journal's requirement, we have archived the underlying analysis code in **Zenodo**, a DOI-assigning repository, under version **1.0.0**.

The repository contains the R and Stata scripts used for data management, machine learning modelling, model evaluation, SHAP-based model interpretation, spatial analysis, and generation of the reported figures and supplementary analyses. The Zenodo DOI has been added to a dedicated **Code Availability** section in the revised manuscript.

Zenodo DOI: [10.5281/zenodo.22712479](https://doi.org/10.5281/zenodo.22712479)

The GitHub repository is also maintained as the accessible source repository for the analysis code.

The GitHub repository is available at <https://github.com/muhammadsalek/SBA-BurkinaFaso-ML-DHS>.

Changes made:

A dedicated **Code Availability** section has been added to the revised manuscript at **Page [11], Lines [296-298]**.

Editor Comment 2: Generation and sources of maps in Figures 6 and 7

Comment:

Please clarify how the maps in Figures 6 and 7 were generated. Please cite the software used to create the maps in the figure legend, including version number and URL link, if the authors created them. Alternatively, if the maps were modified from another source, please cite the sources and provide the required permissions.

Response:

We thank the Editor for this clarification. **Figures 6 and 7 were created by the authors and were not reproduced or modified from previously published maps.** The maps were generated using **R version 4.5.1**, with the **sf** and **ggplot2** packages for spatial data handling and visualization. The maps were produced from publicly available administrative boundary shapefiles and the study data.

Figures 6 and 7 were created by the authors using R version 4.5.1 with the **sf** and **ggplot2** packages. The administrative boundary shapefiles were obtained from the Humanitarian Data Exchange (HDX) Common Operational Datasets (COD) for Burkina Faso and are publicly available at

<https://data.humdata.org/dataset/cod-ab-bfa>.

To improve transparency and reproducibility, we have added the software name, version, relevant packages, and the source of the administrative boundary shapefiles to the respective figure legends. The corresponding mapping code has also been included in the publicly available code repository and archived in Zenodo. Because Figures 6 and 7 are author-generated maps rather than modified versions of previously published figures, no third-party figure copyright permission was required.

Changes made:

The legends of **Figures 6 and 7** have been revised to specify:

- that the maps were generated by the authors;
- R version **4.5.1**;
- the **sf** and **ggplot2** packages;
- the source of the administrative boundary shapefiles; and

- the relevant source URL.

The revised figure legends appear at **Page [31], Lines [768-769] for Figure 6** and **Page [32], Lines [781-782] for Figure 7**.

Summary of revisions

In summary, we have:

1. The underlying analysis code has been made publicly available through the GitHub repository and permanently archived in Zenodo under DOI **10.5281/zenodo.22712479**. The GitHub repository is available at <https://github.com/muhammadsalek/SBA-BurkinaFaso-ML-DHS>.
2. Added a dedicated **Code Availability** section linking the archived code.
3. Clarified the generation of **Figures 6 and 7**.
4. Added the software name and **R version 4.5.1** to the figure legends.
5. Added the relevant spatial packages and the source and URL of the administrative boundary shapefiles to the figure legends.
6. Clarified that Figures 6 and 7 were **created by the authors** and were not reproduced or modified from previously published maps.
7. Ensured that the relevant mapping code is included in the archived reproducibility materials.

We sincerely appreciate the Editor's constructive guidance, which has helped us improve the transparency and reproducibility of the manuscript.

Sincerely,

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